



Integration of process knowledge and statistical learning for the Dow data challenge problem

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ABSTRACT

In this paper, we propose a statistical learning procedure that integrates process knowledge for the Dow data challenge problem presented in Braun et al. (2020). The task is to build an accurate inferential sensor model to predict the impurity in the product stream with apparent drifts. The proposed method consists of i) process data exploratory analysis, ii) a method for variable selection, iii) a method to deal with non-negative physical property modeling using a softplus function; and iv) a method for online bias updating based on known data. We make use of process operation knowledge in all steps of data analytics, including exploratory analysis and feature selection. We report the detection of equipment-switching operations in the data and interpolations found in the impurity data. Partial least squares (PLS) and least angle regression solution (LARS) are adopted to model the data with strong collinearity. Pros and cons of LARS and PLS are given with practical implications.

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1. Introduction

Machine learning and data analytics have taken a center stage in the last decade for engineering and scientific developments. Inspired by human intelligence and learning ability, the recent machine learning advances have been dominated by deep learning led by Hinton and Salakhutdinov (2006) and its successes in i) image recognition and processing; ii) natural language processing (NLP); iii) multilingual translations; iv) competitive games such as AlphaGo, v) and the possibility of autonomous systems such as self-driving cars. Parallel to the deep learning advances are statistical machine learning methods, e.g., Hastie et al. (2009, 2015), which seek not only accurate predictions, but also interpretations and inference with estimates of the error bounds.

Following the tremendous successes of machine learning in the aforementioned areas, manufacturing, operations, and process industries begin to harvest the potential benefits of these breakthroughs (Qin and Chiang, 2019). The chemical industry, among other manufacturing segments, is undergoing a significant transformation guided by the push for Industry 4.0 or smart manu-

facturing. The drive of this industrial revolution is towards advanced data analytics for improved decision making. Although data analytics have been a main area of process systems engineering, e.g., in MacGregor et al. (1994); Chiang et al. (2001); Qin (2003); Venkatasubramanian et al. (2003), the recent interest has been inspired by big data analytics (Qin, 2014; Chiang et al., 2017) to derive insight from diverse data sources.

It has been recognized that domain knowledge is indispensable to develop meaningful and interpretable applications (Qin, 2014; Qin and Chiang, 2019; Chiang et al., 2017). In process systems engineering, Qin and Chiang (2019) state that, *if people do not understand what machine learning is doing, they will not take the risk to trust it*. On the other hand, Ribeiro et al. (2016) in computer vision point out that *if the users do not trust a model or a prediction, they will not use it*. Therefore, interpretability and trustworthiness of machine learning models are prerequisites for them to be adopted in practice.

Amid the successes in image processing, natural language processing (NLP), etc., two kinds of easy efforts have frequently appeared in the literature. One tendency is to indiscriminately apply methods developed in other domains to process engineering problems, such as deep convolutional neural networks (CNN), deep nonlinear auto-encoder-decoder, and long short-term mem-

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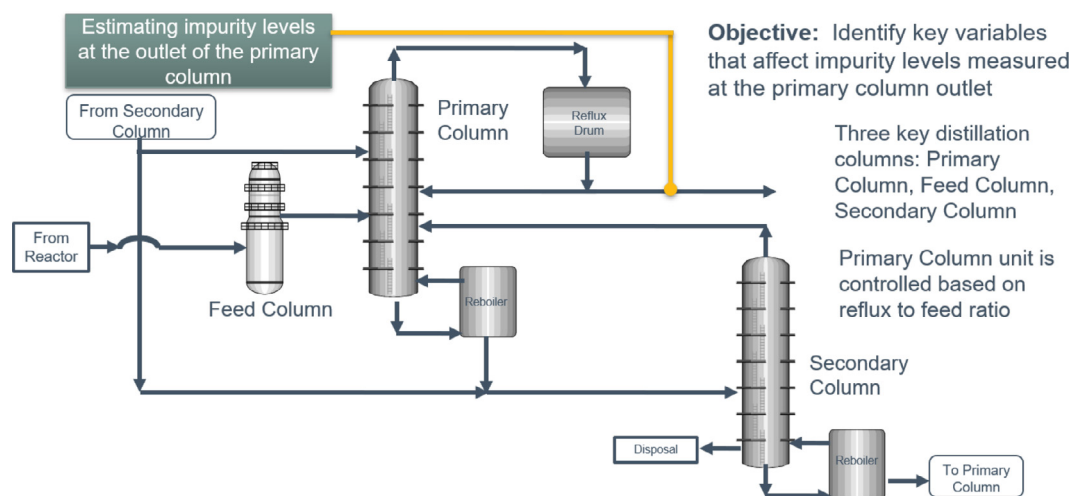


Fig. 1. Block diagram of the Dow data challenge problem considered for the industrial data analytics study.

ory (LSTM) recurrent neural networks. Another tendency is to claim results with marginally improved prediction accuracy based on a real or simulated dataset, but without relevance to process knowledge or interpretation. These tendencies ignore the fact that the successful methods in other domains, such as CNN for image recognition, were developed from a sound understanding of their problems (Fukushima, 1980). They also ignore the fact that the successful methods in image recognition and NLP, etc., have been tested on massive databases. In control and systems, recent benchmark studies on LSTM vs. traditional system identification (Ljung et al., 2020) and recurrent neural networks vs. subspace identification (Sun and Braatz, 2021) show that deep neural net methods do not necessarily outperform traditional methods for long range predictions.

To address the aforementioned shortcomings in process data analytics, this paper focuses on an industrial challenge dataset from Dow, Inc. and propose a solution that integrates process operation knowledge with statistical learning tools. First, process operation knowledge is used in nearly all steps of the analysis, including, i) exploratory data analysis, ii) data sample selection and pre-processing, iii) variable selection, iv) model estimation, and v) prediction validation and interpretation. Second, we rely on statistical learning methods including least angle regression (LARS, [Efron et al., 2004](#)) and partial least squares (PLS, [Hoskuldsson, 1988](#)) for variable selection and interpretability. Third, to guarantee positiveness of the predicted impurity while avoiding unnecessary nonlinear distortion, we propose the use of a softplus function to transform the impurity data, which is a smooth analog to the rectified linear unit (ReLU) popularly used in neural networks. Fourth, to combat the drifting nature of the impurity, we introduce online mean adaptation to the predictive model, which uses available data to update the intercept of the model. The remaining sections of this paper are organized as follows. [Section 2](#) presents the Dow data challenge problem. [Section 3](#) discusses the pre-analysis measures for the Dow data. The proposed integration of process knowledge and statistical learning is given in [Section 4](#). [Sections 5](#) and [6](#) give the detailed results and discussions of the proposed solutions. The final section concludes the paper.

2. The Ddow data challenge problem

Braun et al., 2020 presented the Dow challenge problem that provided masked datasets originating from one of Dow’s processes. In the process section, which is shown in Fig. 1, impurity accumu-

lation resulted in accelerated catalyst aging. The impurity concentration at the Primary Column overhead is measured by a lab testing procedure. The main objective is to predict this lab measured impurity from more than 40 process variables shown in Table 1.

The motivation of monitoring the impurity levels can be shown in Fig. 2. The operation of the refining system, which is the distillation columns shown in Fig. 1, becomes limited when the impurity values from the reactor increase, forcing the refining system to operate sub-optimally in order to remove impurity. Furthermore, Dow's customers enforce specification limits. If the specifications exceed limits, the plant cannot sell the product. Finally, a portion of the outlet of the refining system is recycled back to the reactor, where higher impurity values accelerate catalyst aging.

Two datasets are provided for this case study, one for training a model, and the other for validating the trained model. The training dataset spans from December 2015 to January 2017 and the validation data covers February till October 2017. Possible tasks for the challenge include

1. Not all process variables are relevant to predicting the impurity; therefore, variable selection is desired.
2. The impurity data in the training dataset, as shown in Fig. 3, clearly shows a drift effect due to catalyst aging. It is desirable to have mechanism to deal with this drifting in the process.
3. The dataset has missing values and contains outliers that need to be identified and treated prior to modeling. Data pre-processing is necessary for the modeling task.
4. For the problem to be solved, the process stability and presence of different operating states need to be analyzed and visualized.
5. Interpretability of the modeling results along with high prediction accuracy is desired for process control practice.

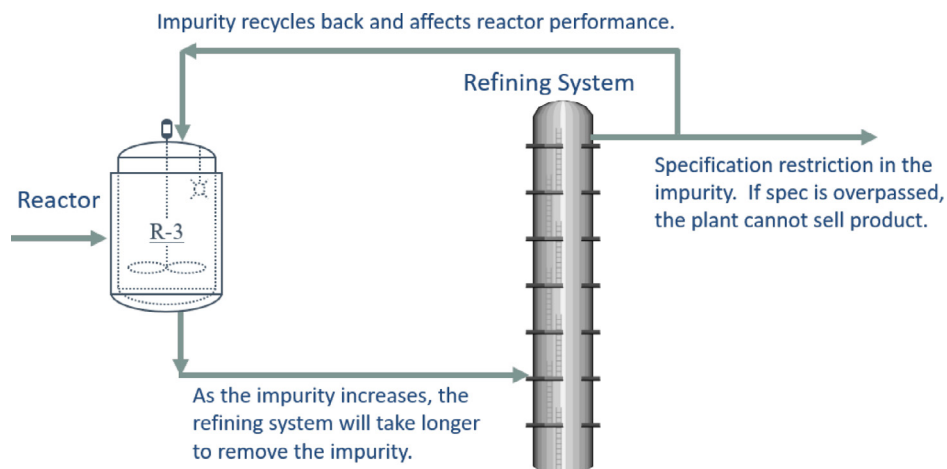
The process data sampling rate is one per hour. There are occasional missing values due to network communication issues. The analyst should decide what to do for the missing values. In this paper, simple preprocessing was conducted before subsequent analysis, including the exclusion of data due to unusual operating changes and elimination of missing or non-operation values. Observations with non-numerical values, such as '!', 'inf', and 'NaN' were removed.

The dataset and detailed instructions can be obtained at <https://dschemaandstats.github.io/dowdatasciencechallenge>. A complete list of tasks are included in the instructions provided alongside the dataset, but the main focus is the development of a reliable and robust prediction model for the impurity concentration. The intention is to make this dataset available to academic

Table 1

The Dow process impurity and more than 40 process variables.

Primary column (PC)	Secondary Column (SC)	Feed Column (FC)
x1:PC Reflux Flow	x22: SC Base Concentration	x36: FC Recycle Flow
x2:PC Tails Flow	x23: Flow from Input to SC	x37: FC Tails Flow to PC
x3:Input to PC Bed 3 Flow	x24: SC Tails Flow	x38: FC Calculated DP
x4:Input to PC Bed 2 Flow	x25: SC Tray DP	x39: FC Steam Flow
x5:PC Feed Flow from FC	x26: SC Head Pressure	x40: FC Tails Flow
x6:PC Make Flow	x27: SC Base Pressure	
x7:PC Base Level	x28: SC Base Temperature	
x8:PC Reflux Drum Pressure	x29: SC Tray 3 Temperature	
x9:PC Condenser Reflux Drum Level	x30: SC Bed 1 Temperature	
x10:PC Bed1 DP	x31: SC Bed 2 Temperature	
x11:PC Bed2 DP	x32: SC Tray 2 Temperature	
x12:PC Bed3 DP	x33: SC Tray 1 Temperature	
x13:PC Bed4 DP	x34: SC Tails Temperature	
x14:PC Base Pressure	x35: SC Tails Concentration	
x15:PC Head Pressure		
x16:PC Tails Temperature		
x17:PC Tails Temperature 1		
x18:PC Bed 4 Temperature		
x19:PC Bed 3 Temperature		
x20:PC Bed 2 Temperature		
x21:PC Bed 1 Temperature		
x41:Avg_Reactor_Outlet_Impurity		
x42:Avg_Delta_Composition PC		
y:Impurity		
x43:PC Reflux/Feed Ratio		
x44:PC Make/Reflux Ratio		

**Fig. 2.** The Dow process refining system recycles impurity back to the reactor, accelerating catalyst aging.

researchers and educators and enable students to study various data analytics approaches on an industrial problem.

3. Data pre-analysis with process knowledge

3.1. Outlier removal

To conduct exploratory analysis of the training data, principal component analysis (PCA) is applied to all input variables. An initial PCA analysis of all input data is shown in Fig. 4 (a), which shows outlying observations compared to the Hotelling's T^2 limit (e.g., MacGregor et al., 1994). Fig. 4 (b) shows that Variable x39 (FC Steam Flow) is the identified main contributor to the outlying observations, which has zero flowrate. These observations are

removed due to the shutdown of the refining system. A subsequent PCA analysis in Fig. 4 (c) shows two distinct clusters in the score space. Further contribution analysis identifies that samples in the red cluster correspond to a large drop in x8 (PC Reflux Drum Pressure). Process operation knowledge confirmed that the operating conditions from 8/22/2016 to 12/16/2016 was changed due to a business decision that will not recur in subsequent operations. Therefore, this period of data will not be used for model training.

3.2. Detecting alternating equipment in operations

The training data sample size after removal of the aforementioned outliers and missing values is about 6159. Fig. 5 shows the

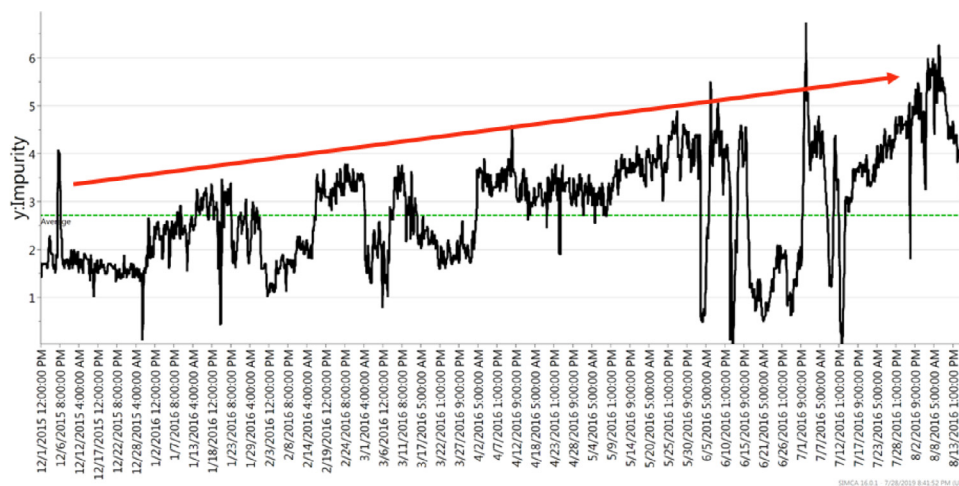


Fig. 3. Drifting is evident in the impurity data of the training dataset.

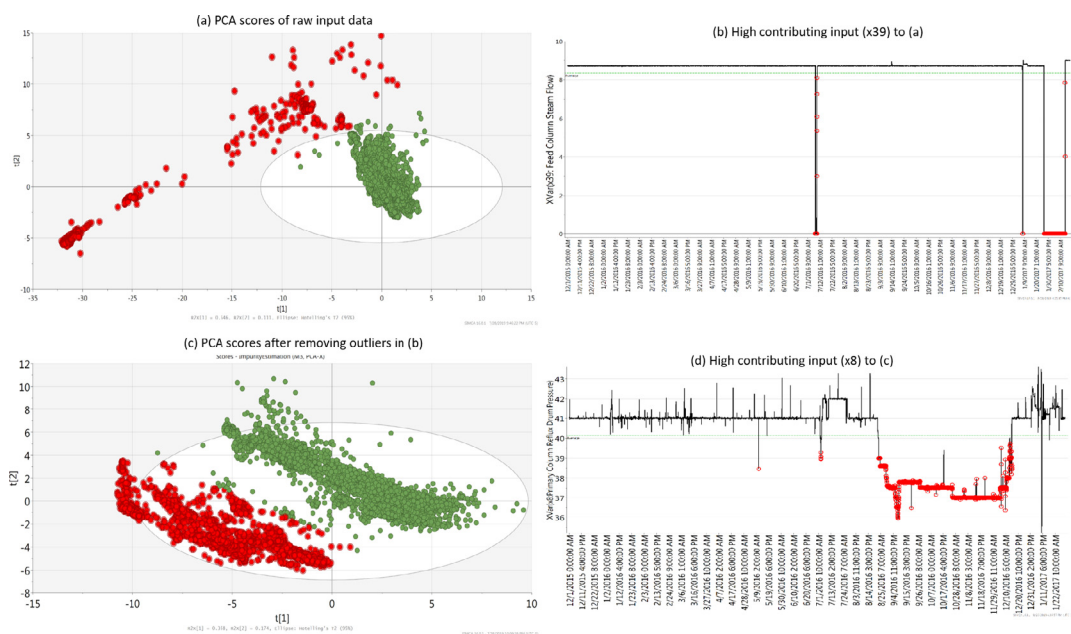


Fig. 4. PCA exploratory analysis of the training data for outlying observations: (a) initial PC scores of all input data shows outlying observations; (b) contribution analysis identifies that x39 (FC Steam Flow) has segments of zero flowrate, which is removed; (c) subsequent PCA analysis shows two clusters in the score space; and (d) contribution analysis identifies that x8 (PC Reflux Drum Pressure) has a large drop.

post-processed training data for the Primary Column, including the impurity data. The alternating colors in the figure are based on the alternating high and low value periods of x16 and x17, highlighted by the dashed green box, which are PC Tail Temperature and PC Tail Temperature 1, respectively.

To further examine the alternating operations, Variables x16 and x17 are re-plotted in the top of Fig. 6. The bottom of Fig. 6 depicts the results of two new variables, with $x16x17H = \max\{x16, x17\}$ and $x16x17L = \min\{x16, x17\}$ for each sample. Surprisingly, the two new trends are continuous and show no mode changes. The trend and range of x16x17L resemble the ambient temperature of a warm weather location with clear seasonal changes. On the other hand, Variable x16x17H forms another continuous trend. It is suspected from these observations that the two PC Tail Temperature sensors are alternately switched on and off, thus measuring the true PC Tail Temperature and ambient temperature alternately. This hypothesis was confirmed by the plant operation personnel. Therefore, Variables x16x17H and x16x17L will

be used for modeling.¹ This simple and proper processing measure avoids the use of complex multi-mode modeling algorithms (e.g., Zhao et al., 2020). It should be noted that such a pre-processing measure would not be found without a sound process understanding.

3.3. Detecting interpolated impurity data points

A close examination of the impurity data reveals that it contains segments of straight lines for several data points. This is evident as shown in Fig. 7, where the top figure shows the neighboring points with circles are on straight lines and the bottom figure is zoomed in for clarity. This observation indicates that the points with circles were linearly interpolated values rather than real measurements. It is further observed that real measurements

¹ The two sensors at the higher modes have a small bias, which has been compensated by equalizing the mean values of data at the higher modes.

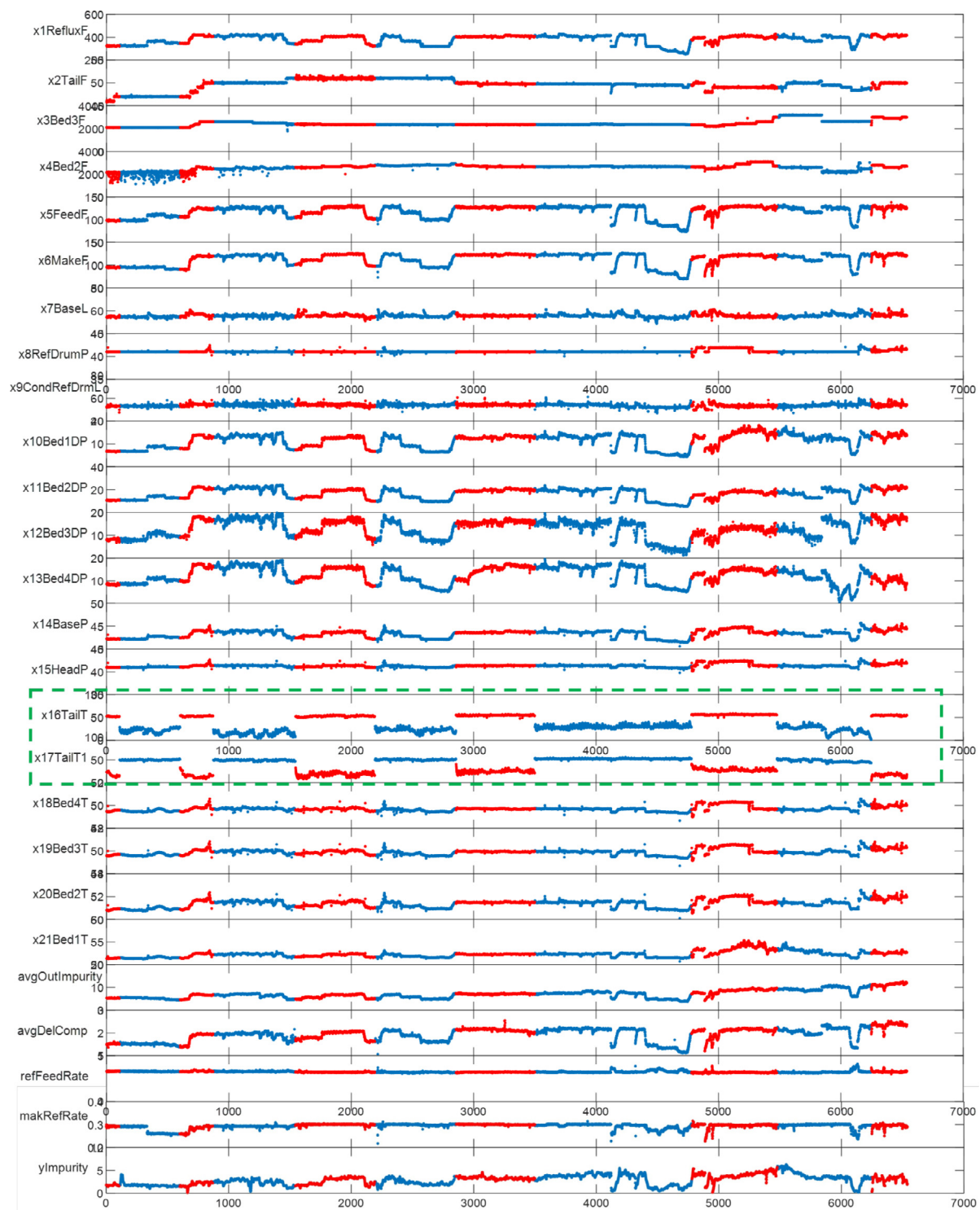


Fig. 5. Training data for the Primary Column. The alternating colors are based on the high and low modes of x16 and x17 within the green dash-lined box. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

were taken at irregular sampling intervals, which is typical for lab-test measurements.

Since the interest is to build a model to predict the impurity, the inter-sample values should be better predicted by the model rather than by linear interpolations. If linear interpolations were any good, it would defeat part of the purpose of building a model. Therefore, it is suggested that interpolated values not be used to train the machine learning models.

3.4. Data transformation for positiveness

When statistical learning models are built to predict the impurity, it is possible that the predictions are negative, which is undesirable. To avoid such results, data transformation is applied. Although a logarithmic transformation would guarantee positiveness by the inverted exponential function, it introduces unnecessary nonlinearity for large positive values of the impurity.

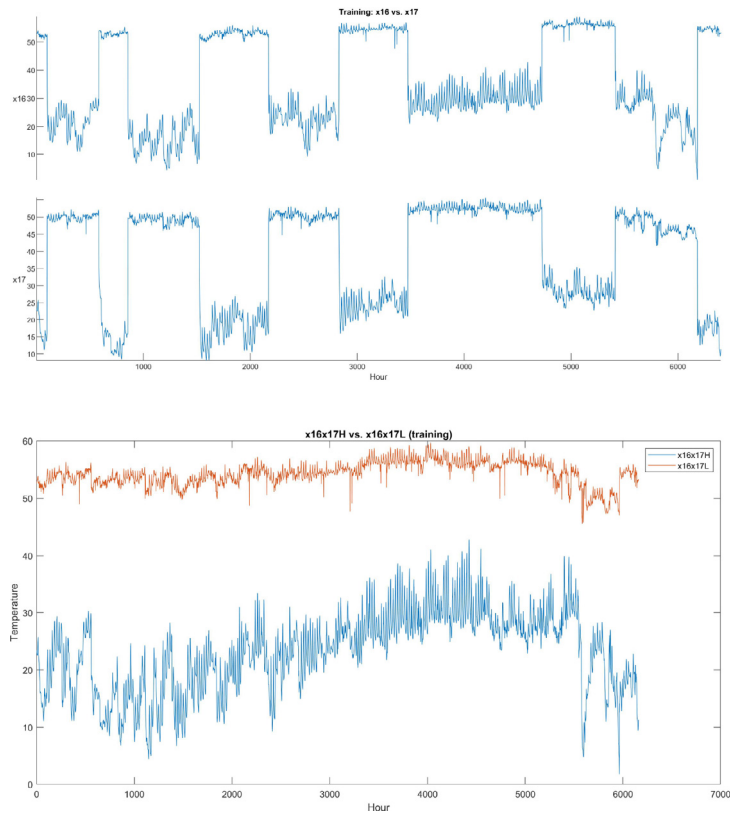


Fig. 6. (Top) Variables x16 and x17 with alternating high and low modes; (Bottom) recombining the higher (and lower) values of x16 and x17 into two new variables x16x17H and x16x17L.

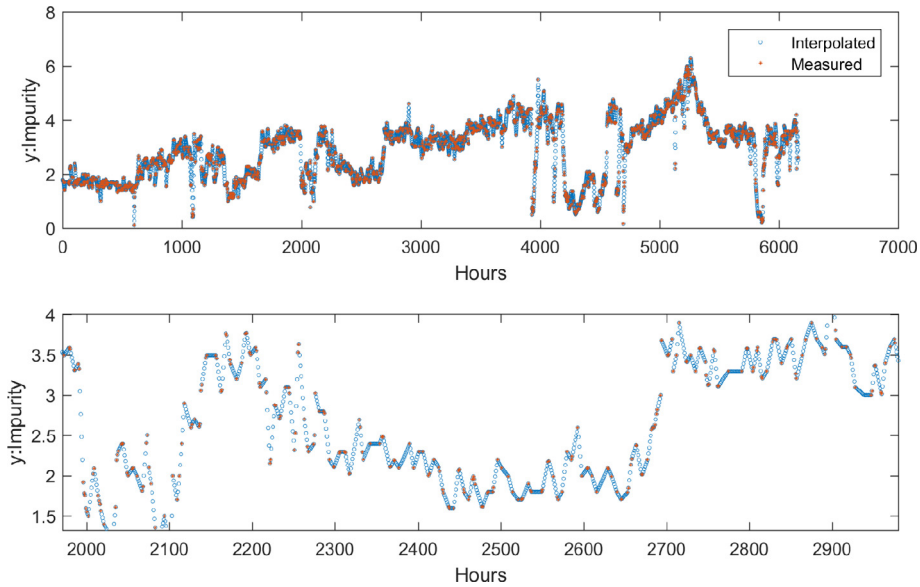


Fig. 7. Impurity data: the top figure shows the neighboring points with circles are on straight lines and points with asterisks are not. The bottom figure is zoomed in.

Inspired from the recent neural network literature, such as (Zheng et al., 2015), we apply the softplus transformation to the impurity data, which has continuous derivatives compared to the rectified linear unit (ReLU) function. The softplus function is depicted in Fig. 8 and is defined as

$$y = \ln(e^z + 1) = s_+(z), \quad (1)$$

where $z = \ln(e^y - 1) \in \mathbb{R}$ is the transformed variable. The transformation guarantees that $y > 0$ no matter what z is. When y is large, the softplus asymptotically approaches $y = z$, which avoids unin-

tended nonlinear distortion of other transformations such as a logarithmic transformation.

4. Proposed method with process knowledge integration

4.1. The proposed method

In the pre-analysis discussed in the previous section, it is demonstrated that process operation knowledge is indispensable in nearly every analysis step. Therefore, the proposed method is

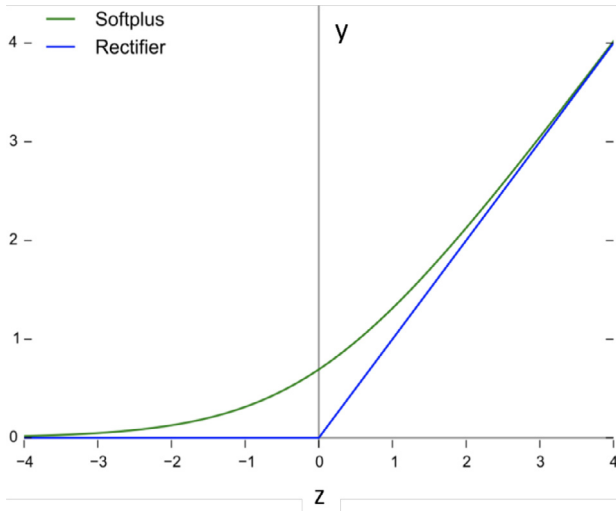


Fig. 8. The softplus (vs. ReLU in blue) transform adopted to guarantee positiveness of the predicted impurity. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

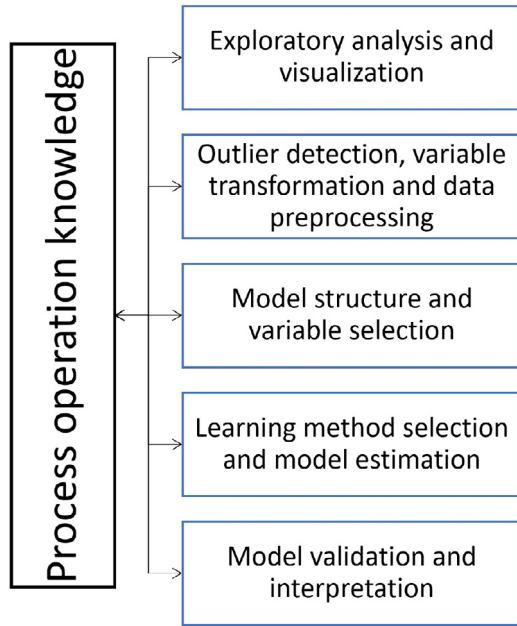


Fig. 9. Flowchart of the interactive application of process operation knowledge in data analysis.

an interactive application of process operation knowledge and data analysis in virtually all steps of modeling. Although the proposed method is problem-driven, it is generally representative of sustainable industrial data analytic applications. The proposed solution reflects decades of data analytics research and experience of the first author and the Dow coauthors' application experience and process understanding.

The proposed interactive application of process operation knowledge and data analysis is depicted in Fig. 9. The bidirectional arrows indicate the interactive nature of knowledge and analysis applications to the process data. The outcomes are also bidirectional in the sense that data analysis and process understanding help each other. Exploratory analysis and visualization is the first step, where principal component analysis can be very helpful in this step. As demonstrated in the previous section, outliers in the dataset and major changes of operations can be easily spot-

ted with this tool. Softplus transformation was performed on the dataset to avoid predicting negative values.

To perform variable selection with collinearity observed in the data, as depicted in Figure 5, PLS and LARS methods are selected to model the data. PLS is robust to collinearity and has good interpretability with large weights indicating high association with the impurity. LARS has the ability to select relevant variables used in the way of the Lasso (Tibshirani, 1996). One purpose of applying LARS to the Dow problem is to test how it deals with collinearity among the predictors that are highly correlated with the response variable. To improve the prediction in the presence of process drifting, a bias correction method with a Kalman filter gain is applied to the models. The resulting models show a much improved prediction accuracy on the validation dataset. Interpretations and validation of the highly relevant variables in both models are conducted based on process knowledge.

4.2. PLS mModeling

The PLS regression method is well documented in the literature. Manne (1987) gave a couple of PLS algorithms that are geometrically equivalent. Rosipal and Krämer (2006) summarized some recent development of several PLS algorithms, including the connection to a conjugate gradient solution to the least squares objective. A major feature of PLS methods is the ability to reduce dimensions by selecting the number of latent factors before reaching the least squares solution. Therefore, the number of latent factors is a tuning parameter to provide regularization in the regression coefficients. Shrinkage is achieved by tuning this parameter (Rosipal and Krämer, 2006), which is different from the methods of LARS and the lasso.

Let $\mathbf{x}_k \in \mathbb{R}^p$ be predictor vector and y_k the response variable, which are scaled to zero mean and unit variance. The vector of parameters $\boldsymbol{\beta}$ that we are trying to estimate satisfies the model

$$y_k = \mathbf{x}_k^T \boldsymbol{\beta} + \varepsilon_k. \quad (2)$$

A data matrix $\mathbf{X}$ is formed with N samples as

$$\mathbf{X} = [\mathbf{x}_1 \ \mathbf{x}_2 \ \dots \ \mathbf{x}_N]^T \in \mathbb{R}^{N \times p}.$$

With response or label data $\mathbf{y} \in \mathbb{R}^N$, supervised latent variable analytics aim to predict $\mathbf{y}$ using latent variables in $\mathbf{X}$. The algorithms of PLS (Geladi and Kowalski, 1986; Hoskuldsson, 1988; Helland, 1988) find a latent structure iteratively to achieve maximized covariance. Initialized with $\mathbf{X}_1 = \mathbf{X}$, $i = 1$, and the respective latent score vector $\mathbf{t}_i = \mathbf{X}_i \mathbf{w}_i$, the PLS objective is expressed as

$$\max_{\mathbf{w}_i} J = \mathbf{w}_i^T \mathbf{X}_i^T \mathbf{y} \quad (3)$$

$$\text{s.t. } \|\mathbf{w}_i\| = 1.$$

To calculate the next latent variable, the residual

$$\mathbf{X}_{i+1} = \mathbf{X}_i - \mathbf{t}_i \mathbf{p}_i^T \quad (4)$$

is calculated to obtain the next factor, where $\mathbf{p}_i = \mathbf{X}_i^T \mathbf{t}_i / (\mathbf{t}_i^T \mathbf{t}_i)$.

It is easy to see that the PLS solution to (3) is $\mathbf{w}_i \propto \mathbf{X}_i^T \mathbf{y}$. Since $\mathbf{t}_i = \mathbf{X}_i \mathbf{w}_i$, it is sometimes preferred to re-scale $\mathbf{t}_i$ to unit norm after $\mathbf{w}_i$ is obtained. With L latent variables, the latent scores $\mathbf{T} = [\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_L]$ are orthogonal (Hoskuldsson, 1988) and we have $\mathbf{T}^T \mathbf{T} = \mathbf{I}$. Defining $\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_L]$ and $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_L]$, we have $\mathbf{T} = \mathbf{X} \mathbf{R}$ and

$$\hat{\mathbf{y}} = \mathbf{T}(\mathbf{T}^T \mathbf{T})^{-1} \mathbf{T}^T \mathbf{y} = \mathbf{T} \mathbf{T}^T \mathbf{y} = \mathbf{X} \mathbf{R} \mathbf{R}^T \mathbf{X}^T \mathbf{y}, \quad (5)$$

where the estimate of the PLS regression coefficients is

$$\hat{\boldsymbol{\beta}}_{pls} = \mathbf{R} \mathbf{R}^T \mathbf{X}^T \mathbf{y}. \quad (6)$$

and $\mathbf{R} = \mathbf{W}(\mathbf{P}^T \mathbf{W})^{-1}$. Comparing to the ordinary least squares solution (OLS), $\hat{\boldsymbol{\beta}}_{ols} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$, PLS avoids inverting the covariance matrix $\mathbf{X}^T \mathbf{X}$.

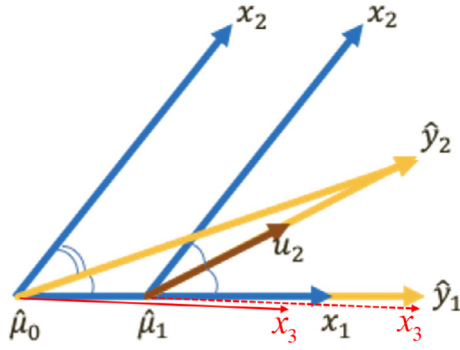


Fig. 10. Illustration of the least angle regression for a two-variable case .

In PLS, the model estimate $\hat{\beta}_{pls}$ is calculated for a number of latent variables. The optimal number of latent variables is often determined using cross validation, aiming to achieve the best prediction.

4.3. LARS mModeling with variable selection

Similar to PLS that provides a sequence of partial models, the LARS procedure by Efron et al. (2004) selects the most relevant predictors to the response variable one by one, until the optimal model is selected with a criterion. Unlike PLS that uses all predictor variables in each of the partial models, LARS adds a new variable to the model if its angle with the current residual becomes as small as the angle between the current model and the residual.

The LARS forward selection process is depicted in Fig. 10. The LARS procedure by Efron et al. (2004) first finds the predictor most correlated with the response. Then the largest step possible in the direction of this predictor is taken until another predictor has as much correlation with the current residual. LARS proceeds in the equi-angular direction between the two predictors until another variable becomes most correlated with the new residual. This process is iterated until the model yields the best prediction measured by the C_p -type risk estimate at each step. Efron et al. (2004) also pointed out that a variant of LARS is equivalent to the Lasso solution (Tibshirani, 1996) popularly used for variable selection.

The LARS algorithm to select variables and estimate coefficients β in (2) can be summarized as follows (Efron et al., 2004).

1. Start with $\beta = 0$. Find the predictor x_j that is most correlated with y .
2. Increase the coefficient β_j in the direction of the same sign as its correlation with y . Calculate residual $r = y - \hat{y}$ along the way. Stop when some other predictor x_l has as much correlation with r as x_j does.
3. Use the active set $\{x_j, x_l\}$ to predict y and increase in the least squares direction until another predictor x_m has as much correlation with the new residual r .
4. Repeat the above procedure until all predictors are included in the model; calculate C_p vs. the number of selected variables and find a knee point to stop selecting additional variables.

Like PLS, LARS approaches the ordinary least squares (OLS) solution if all p variables are included in the model. The C_p criterion selects a subset of variables in the model and effectively makes other variables zero coefficients. However, the LARS model with $m < p$ variables is not the same as the OLS solution with the m variables. It backs off from the OLS solution with m predictors until its residual reaches equi-angle with the $(m+1)^{th}$ variable. Therefore, the $(m+1)^{th}$ variable plays a role in the LARS model, even though only m variables are selected in the model.

Unlike PLS that is designed to deal with collinearity explicitly with latent projections, the LARS's ability to deal with collinearity has not been much explored. In essence, LARS is a greedy approach that picks the next variable with equal correlation to the residual of the already selected variables. Collinear variables in a subset tend to be represented by one selected variable in the subset while others will likely be excluded. This can be illustrated with Variable x_3 in Fig. 10, which is collinear with Variable x_1 that has been selected. As LARS proceeds along the direction of x_1 , the angle between the residual and x_3 will increase. This will prohibit x_3 to be selected, even though it is highly correlated with the response. In this study, we will show the impact of collinearity on LARS using the Dow dataset with strong collinearity.

While LARS is designed to select variables sequentially which can yield the Lasso forward selection path, Lasso has been shown to give non-robust variable selection results due to the use of l_1 norm (Meinshausen and Bühlmann, 2010). Elastic nets proposed by Zou and Hastie (2005) help alleviate this issue by using a mixed l_1/l_2 norm. Although these methods are different from the forward selection in linear regression as demonstrated by Hastie et al. (2009), there has been recent work in the context of many predictors where forward stepwise selection methods are not recommended (Smith, 2018).

The similarity and differences of LARS and PLS can be illustrated geometrically in Fig. 11. Both methods build models sequentially with one dimension at a time. Both methods approach the OLS solution when all variables are included. Further, both methods implement non-orthogonal projections when a subset of variables is selected. Their differences can be summarized as follows.

1. While LARS picks the winner with the least angle as the next variable to include, PLS includes all variables in each dimension weighted by the covariance with the response.
2. While LARS stops short of the orthogonal projection to an active set of variables when the next equi-angle competitor is encountered, PLS reaches the orthogonal projection to the included latent dimensions.

4.4. Process drift adaptation

Since the process is known to drift over time due to catalyst aging, no fixed models are expected to do well without some adaptation to the drift. A simple approach is to update the mean of the model with a conditional mean on recently available data, while keeping the model parameters unchanged. This method is referred to as online bias learning (OBL) which adapts for the bias between online drifting process data and model predictions. Online bias learning has been a common practice in adapting for drift disturbances in run-to-run control (e.g., Wang et al., 2005).

Since the output data are measured at irregular sampling intervals, we use $k(i)$ to denote the current time and $y_{k(i-j)}$, $j = 1, 2, \dots, n$ the most recent data available to use for the bias adaptation. Let $f(\mathbf{x}_{k(i)})$ be the current model prediction without bias adaptation. The process with mean drift $d_{k(i)}$ can be expressed as

$$y_{k(i)} = f(\mathbf{x}_{k(i)}) + d_{k(i)} + v_{k(i)}, \quad (7)$$

where $v_{k(i)}$ is random noise with zero mean. We use the n available observations to estimate $d_{k(i)}$ as follows,

$$\hat{d}_{k(i)} = \frac{1}{n} \sum_{j=1}^n (y_{k(i-j)} - f(\mathbf{x}_{k(i-j)})).$$

Noticing that $\hat{d}_{k(i+1)} = \frac{1}{n+1} \sum_{j=0}^n (y_{k(i-j)} - f(\mathbf{x}_{k(i-j)}))$ when $y_{k(i)}$ is available, we have

$$(n+1)\hat{d}_{k(i+1)} = n\hat{d}_{k(i)} + (y_{k(i)} - f(\mathbf{x}_{k(i)})) = (n+1)\hat{d}_{k(i)} + (y_{k(i)} - \hat{y}_{k(i)})$$

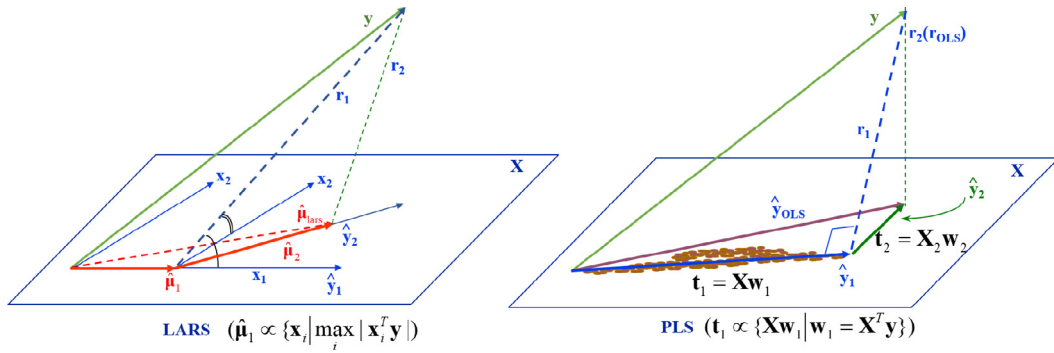


Fig. 11. Geometric illustration of LARS vs. PLS in searching for variables or latent directions for regression .

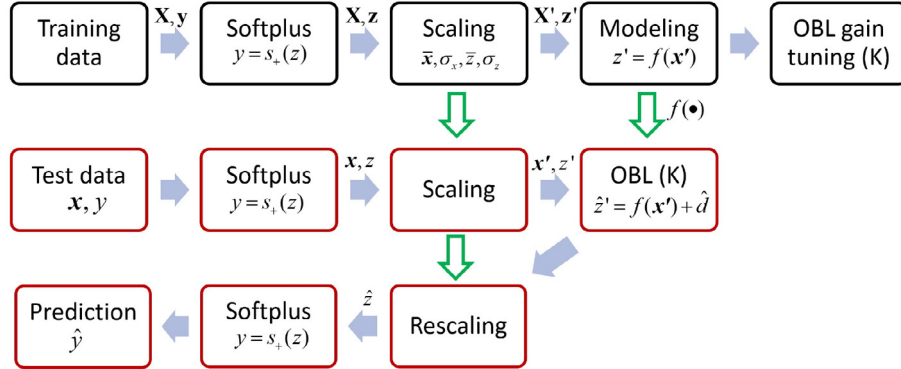


Fig. 12. Flowchart for data transformation, scaling, and OBL updating: the first row for model building and the last two rows for prediction with the OBL update.

which yields the OBL formula,

$$\hat{d}_{k(i+1)} = \hat{d}_{k(i)} + K(y_{k(i)} - \hat{y}_{k(i)}), \quad (8)$$

where

$$\hat{y}_{k(i)} = f(\mathbf{x}_{k(i)}) + \hat{d}_{k(i)} \quad (9)$$

and $K = 1/(n+1)$.

The tuning parameter K has a Kalman filter interpretation which ranges from 0 to 1. If K is larger, the bias term depends more on the most current residual. The OBL is analogous to an exponentially weighted moving average (EWMA), except that the sampling interval is irregular. In practice, K can be tuned based on the training data, where the effect of irregular sampling intervals is representative. If the lab-test sampling interval changes significantly, K should be re-tuned.

4.5. Integration of data transformation, model building and drifting adaptation

With the proposed problem-driven method to build a reliable model for prediction with drift adaptation, it is important to align the modeling tasks in the right sequence. Fig. 12 depicts that appropriate sequence for data transformation, scaling, and OBL updating. It is important to note the following.

- The softplus transform is done on the raw data y_k to obtain z_k , then the transformed variable will be scaled to zero mean and unit variance. The model of the scaled variables should have a zero intercept.
- The OBL update in the figure is performed on scaled data. An alternative is to perform it on the re-scaled data. However, if the OBL is performed on the raw data scale y_k , it would still be possible for $\hat{y}_k$ to be negative.

5. LARS and PLS analysis results

5.1. Model benchmark indices

To compare the effectiveness of models derived from the dataset, two indices, R^2 for the training data and Q^2 for test data (or out of sample data) are used for evaluation of a model on training and test data. The R^2 index is a measure of a model in predicting the training data after properly selecting the model hyper-parameters. The calculation of R^2 is

$$R^2 = 1 - \frac{\sum_{k \in \text{trn}} (y_k - \hat{y}_k)^2}{\sum_{k \in \text{trn}} (y_k - \bar{y}_{\text{trn}})^2}, \quad (10)$$

where $\bar{y}_{\text{trn}}$ is the mean of the response calculated from the training data. Note that the R^2 depends on the optimally determined hyper-parameters that provide a trade-off between model bias and variance, such as the number of PLS factors or the selected variables by LARS.

The Q^2 index is analogous to R^2 by applying the model to a validation dataset or an out-of-sample dataset,

$$Q^2 = 1 - \frac{\sum_{k \in \text{val}} (y_k - \hat{y}_k)^2}{\sum_{k \in \text{val}} (y_k - \bar{y}_{\text{trn}})^2}. \quad (11)$$

Note that the mean $\bar{y}_{\text{trn}}$ is derived from the training data. One cannot replace $\bar{y}_{\text{trn}}$ with $\bar{y}_{\text{test}}$ since $\bar{y}_{\text{trn}}$ is part of the trained model.

If above definition of Q^2 in (11) is applied to cross-validation, the numerator of the last term would be the predicted error sum of squares (PRESS) in Wold et al., 2001. In this work, we have a truly out-of-sample set to validate the model, which is not used for model training at all. In the sequel we present the modeling and prediction results of PLS and LARS with several scenarios.

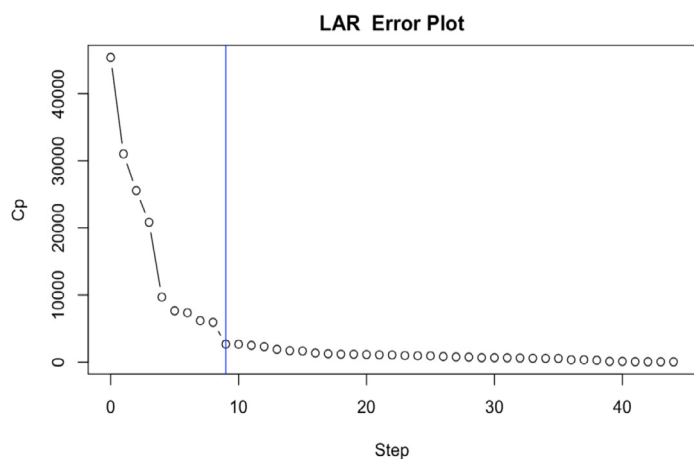


Fig. 13. C_p -risk estimates vs. the number of variables included in the model.

5.2. LARS rResults

In remaining part of this section, we report the model learning and prediction results on the training set as well as the validation set, which is kept away from model training and parameter tuning. The results are derived under the conditions that i) the pre-processing measures in Section 3 have been applied to the data; ii) softplus is applied to guarantee positive predictions of the impurity; and iii) only real measured impurity data are used for model training and validation; interpolated data are excluded. Comparison of the modeling results trained with and without the interpolated values will be conducted for empirical evidence. However, it would be inappropriate to use interpolated label data to validate a model.

For LARS, the C_p -risk estimation is used to assess the error and select relevant variables to retain in the model. Fig. 13 shows C_p -risk estimates vs. the number of variables included in the model. It is seen that the inclusion of nine variables will bring the C_p to a 'knee' point.

By using the first nine variables, the model fitting result on the training dataset and the test result on the validation dataset are shown in Fig. 14. The model yields fairly good result on the training dataset with a R^2 value of 0.825, whereas the Q^2 value on the validation set is 0.681. The bottom trend chart Fig. 14 shows a bias, which is expected since the process drifts over time.

To adapt for the bias with the LARS model, OBL is applied to the training dataset and the validation dataset. The OBL gain K is tuned to be 0.74 with the training dataset and the results on both training and validation sets are shown in Fig. 15. It is seen that the R^2 value is improved from 0.825 to 0.923, while the Q^2 value on the validation set is improved from 0.681 to 0.882. The results show that, conditioned on the recently measured impurity data, the OBL is effective in compensating for process drift without adapting the LARS model coefficients.

5.3. PLS results

Although PLS does not perform variable selection, it handles collinearity and provides shrinkage in regression with the number of latent variables selected by cross-validation. Larger PLS coefficients indicate that the associated variables are more important in predicting the response. To determine the optimal number of components, the training dataset is divided into 10 folds. Then, 10 trials of model fitting are conducted. In each trial, nine of these folds are used for training while the remaining one for validation to calculate the predicted mean square error (MSE). After the rotation, the optimal number of components that bring the best overall MSE is

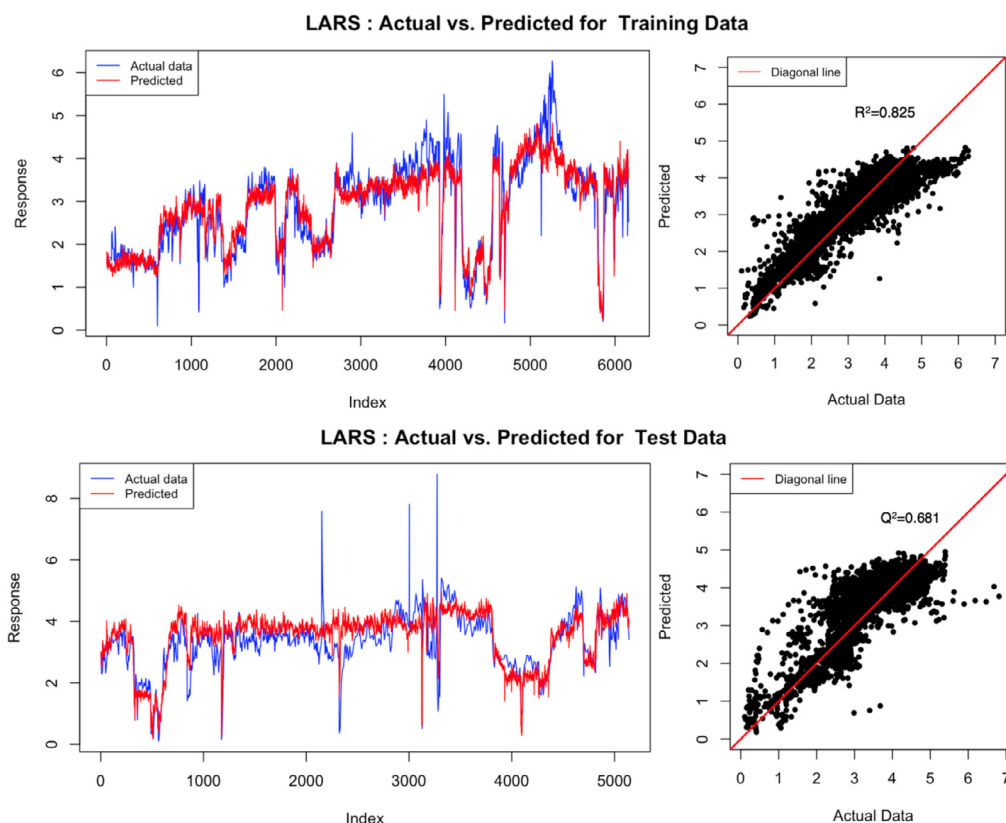


Fig. 14. The LARS model fitting result on the training dataset (Top) and the testing result on the validation dataset (Bottom).

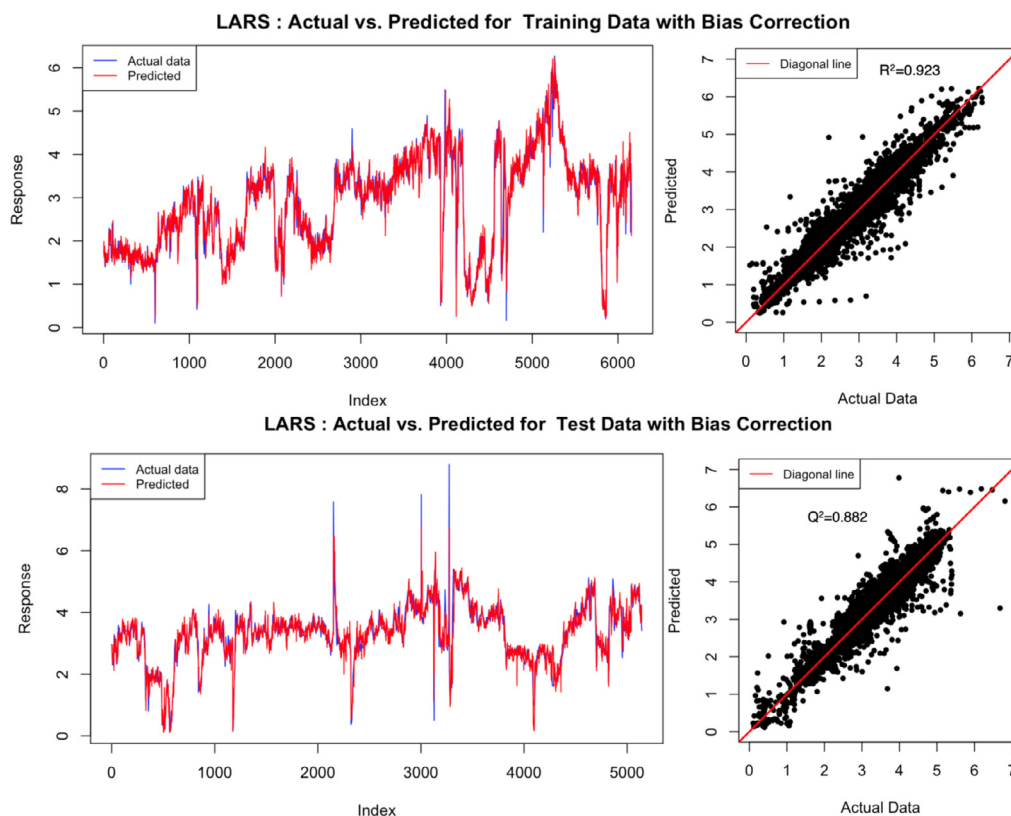


Fig. 15. The LARS model predictions with OBL updates on the training dataset (Top) and the validation dataset (Bottom).

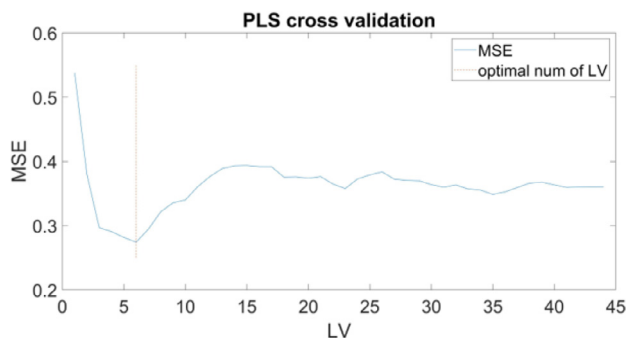


Fig. 16. Cross-validation errors vs the number of latent variables in the PLS model.

selected. As shown in Fig. 16, six components produce the best result.

A final PLS model with six components is fitted with all observations in the training dataset. The fitting result is shown in the top of Fig. 17. The test result on the validation dataset is shown in the bottom of Fig. 17. It is evident that while the fitting result on the training dataset achieves a R^2 value of 0.863, the Q^2 value on the validation set is only 0.508. The bottom trend chart Fig. 17 shows a clear bias, which is expected since the process drifts over time.

To adapt for the process drift, OBL is used to improve the PLS predictions. The OBL gain K is tuned optimally to be 0.83 with the training dataset. The OBL prediction results with the PLS model on the training dataset and the validation dataset are shown in Fig. 18. We can see that the R^2 value is improved from 0.863 to 0.943, while the Q^2 value on the validation set is improved from 0.508 to 0.880. Comparing to the OBL results from the LARS model, OBL

with the PLS model yields marginally better R^2 value but virtually the same Q^2 value on the validation dataset.

6. Discussions and comparisons

6.1. Interpreting the LARS and PLS models

Although the LARS and PLS models are comparable in terms of the prediction accuracy, LARS provides a succinct list of nine variables, while PLS uses all variables and projects them to six latent variables. With all variables, PLS is more capable in fitting the data than LARS with nine variables. However, this difference is adequately compensated by the OBL correction on the validation set.

To interpret the two models, their regression coefficients are shown in Fig. 19, where the LARS-selected variables are rank-ordered first, followed by the remaining variables rank-ordered by the magnitudes of the PLS coefficients. Since the predictors are scaled to unit variance, the magnitudes of the coefficients indicate of the relative importance of the predictors. Although variables with tiny PLS coefficients were not explicitly deleted in the PLS model, their impact on the model is outweighed by variables with much larger coefficients. The sequence of variables included in the LARS model is shown on the left side of the figure. It is seen that variables with tiny PLS coefficients appear rather late in the LARS sequence.

It is observed that the 11 variables with the largest PLS coefficients, highlighted with the green box in Fig. 19, include all nine variables selected by LARS. Two variables, x2:PC Tails Flow and x6:PC Make Flow, have large PLS coefficients in magnitudes, but they are not selected by LARS. Instead, LARS selects x44:PC Make/Reflux Ratio, which takes the place of x6:PC Make Flow. Since x44 is a calculated variable, it can be regarded as a transformed variable using process knowledge.

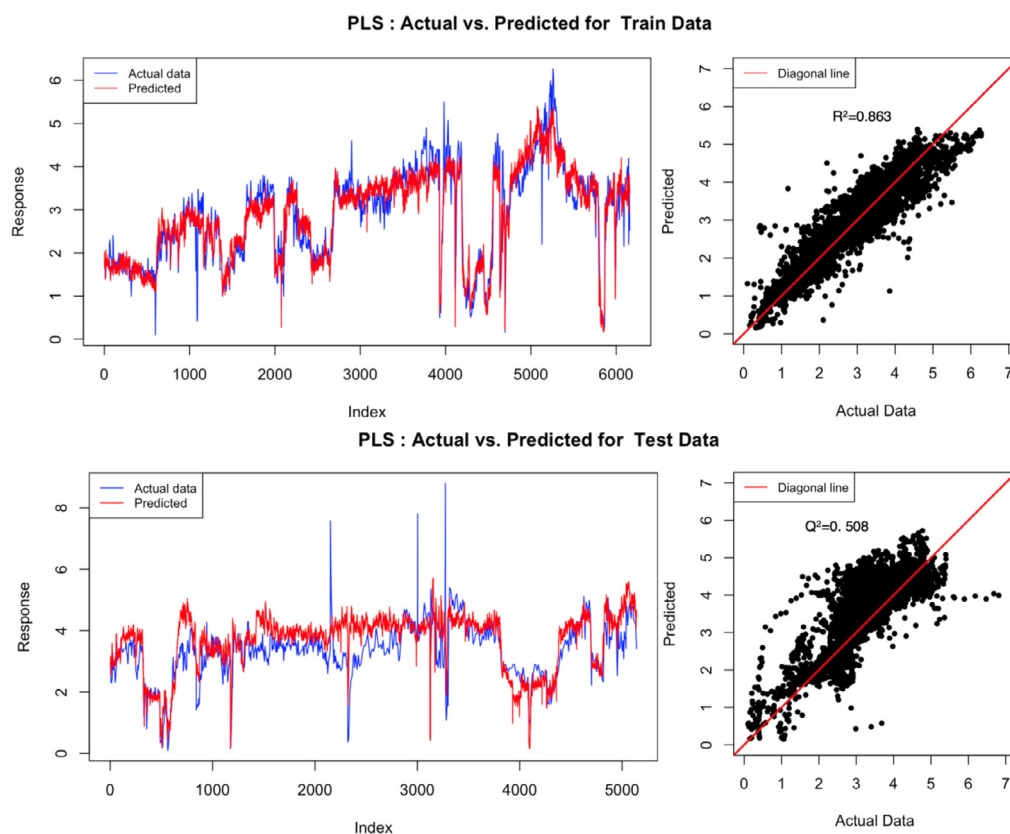


Fig. 17. The PLS model fitting result on the training dataset (Top) and the testing result on the validation dataset (Bottom).

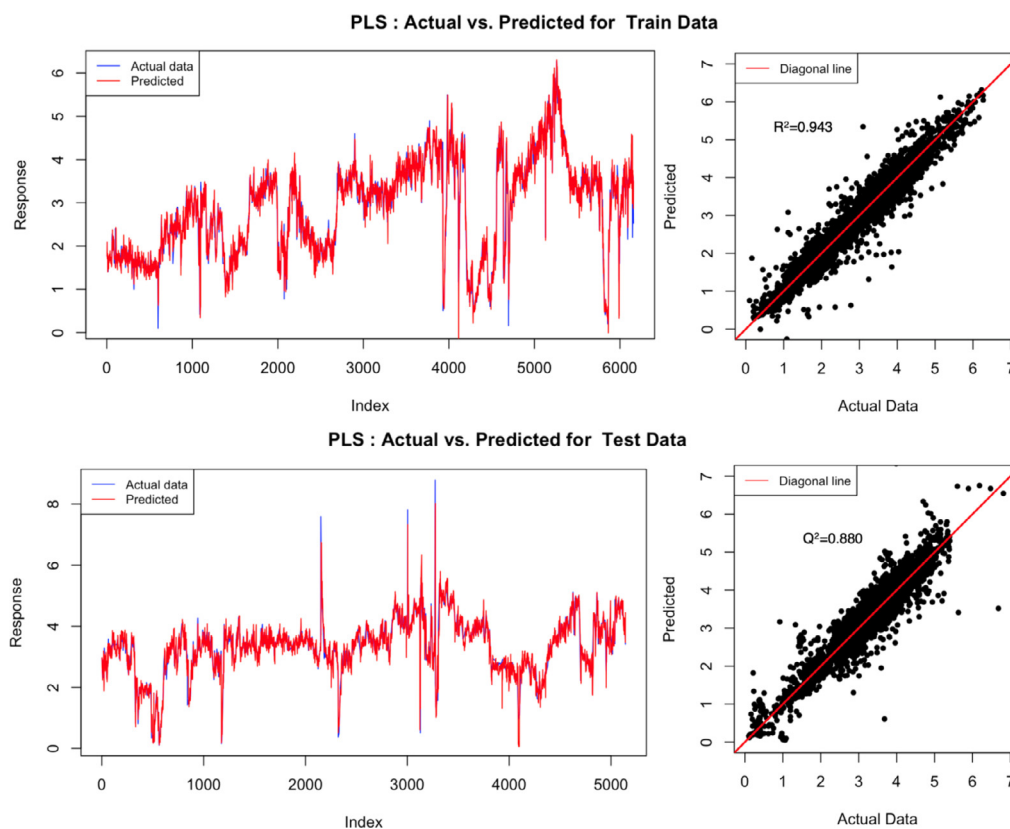


Fig. 18. The PLS model predictions with OBL updates on the training dataset (Top) and the validation dataset (Bottom).

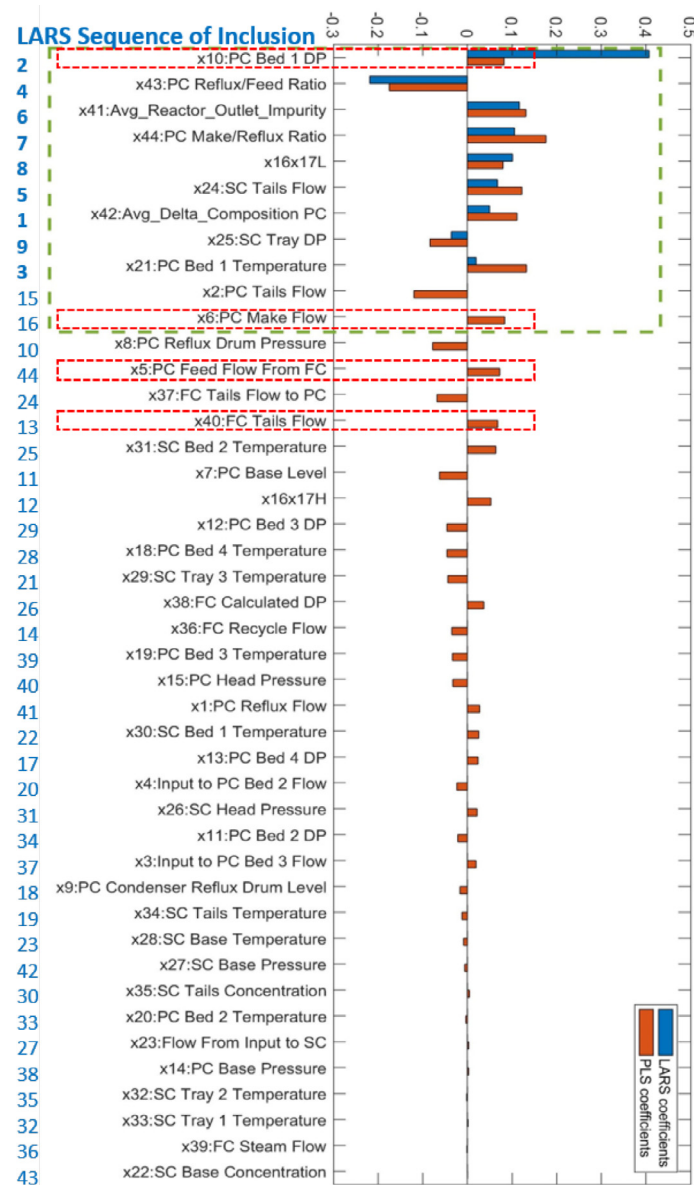


Fig. 19. The LARS selected variables are also large-magnitude coefficients in the PLS model. The top 11 PLS variables include all nine variables picked by LARS.

It is interesting to observe that none of the variables in the Feed Column are selected by LARS. Two variables are selected from the Secondary Column, which are SC Tray Delta Pressure (x25) and SC Tails Flow (x24), while the rest of the LARS selected variables are from the Primary Column. The PC Bed 1 Delta Pressure and Temperature as well as the Reactor Outlet Impurity are selected, which makes sense according to process knowledge. Finally, the PC Tail Temperature (x16x17H) is not selected by LARS, but the ambient temperature (x16x17L) is selected with a positive coefficient. This is not surprising since ambient temperature usually works as a disturbance to the plant operations.

One remaining question is how LARS treats collinear variables that are also highly relevant to predicting the Impurity. To check the collinearity of the predictors we use hierarchical clustering (James et al., 2013) to find subsets of variables with high collinearity. We calculate the correlations r_{ij} among them and use $(1 - r_{ij}^2)$ to measure the dissimilarity between Variables x_i and x_j . The hierarchical clustering results are depicted in Fig. 20, where the complete linkage criterion is applied (James et al., 2013). The vertical line in Fig. 20 corresponds to $|r_{ij}| = 0.9$, which identifies clusters

with high collinearity. With the complete linkage we guarantee that no two variables in each cluster have a correlation $|r_{ij}| < 0.9$.

Table 2 depicts the nine variables selected by LARS, four variables in the same collinear cluster with x10 (Bottom) by the clustering analysis in Fig. 20, and their correlations with the Impurity and x10. It is seen from Fig. 20 that none of the LARS-selected variables are in the same cluster with a correlation $|r_{ij}| > 0.9$. Therefore, LARS does not select more than one variable from a collinear subset. For example, Variable x10 has the largest coefficient in the LARS model. It shares the same cluster as Variables x6, x5, x40, and x1, but LARS did not select any of these variables.

Within the cluster of Variables x10, x6, x5, x40, and x1, the correlations in Table 2 show that they have high correlations with the Impurity, but only Variable x10 is selected by LARS. By examining Fig. 19, Variables x6, x5, and x40, highlighted in red boxes, have large PLS coefficients with nearly the same magnitude as Variable x10 does. Therefore, PLS spreads the weights among these collinear variables, while LARS relies on Variable x10 only, which is a greedy result.

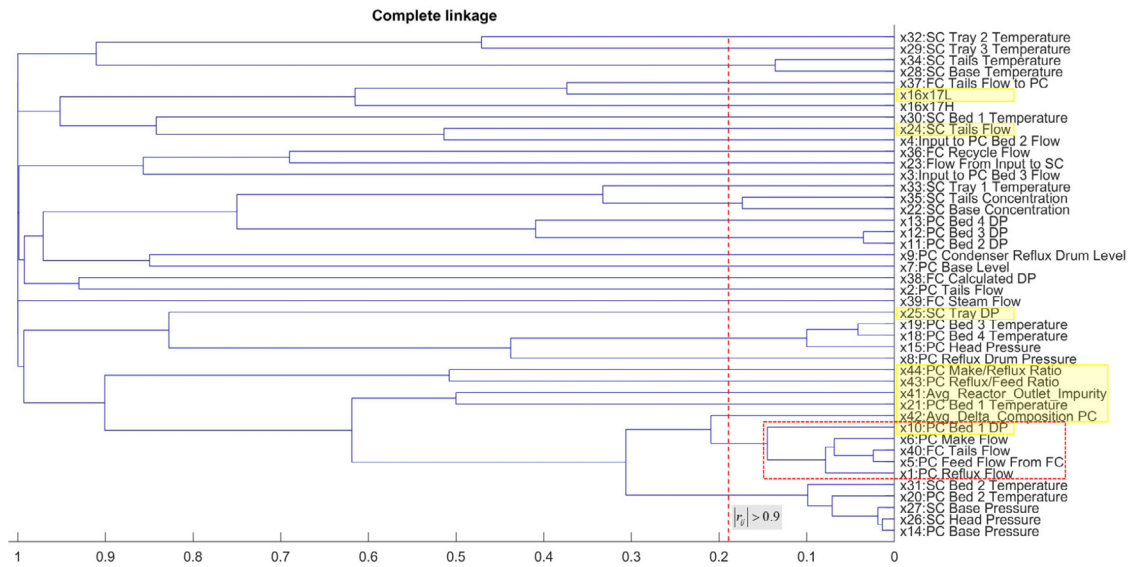


Fig. 20. Hierarchical clustering finds subsets of variables with high collinearity. The dashed vertical line corresponds to $|r_{ij}| = 0.9$. The red rectangle highlights variables collinear with x10. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2

The nine LARS-selected variables (Top), four variables collinear with x10 (Bottom), and their correlations with the Impurity and x10.

Predictors selected by LARS	y:Impurity	x10
x10:PC Bed 1 DP	0.79	–
x43:PC Reflux/Feed Ratio	–0.61	–0.38
x41:Avg Reactor Outlet	0.73	0.74
x44:PC Make/Reflux Ratio	0.61	0.46
x16x17L:Ambient Temperature	0.33	0.04
x24:SC Tails Flow	0.59	0.41
x42:Avg Delta Composition PC	0.81	0.89
x25:SC Tray DP	–0.51	–0.38
x21:PC Bed 1 Temperature	0.75	0.80
Predictors collinear with x10		
x6:PC Make Flow	0.76	0.92
x5:PC Feed Flow from FC	0.77	0.94
x40:FC Tails Flow	0.78	0.94
x1:PC Reflux Flow	0.69	0.94

The question is, is it better to select one variable in a collinear set or to include all collinear variables that are strong predictors for the output? The answer depends on how the model will be used. For the purpose of predictions, either approach could work well. If the purpose is to control the output by manipulating the model input, the LARS model would manipulate Variable x10 only and assume Variables x6, x5, and x40 would not change. However, this would violate the process collinearity among these variables, since it is not possible to change x10 (PC Bed1 DP) without changing x6 (PC Make Flow) and x5 (PC Feed Flow).

Finally, Variable x1:PC Reflux Flow in this collinear group is not important in either model. The PC Reflux Flow is represented with two calculated variables, x43 (PC Reflux/Feed Ratio) and x44 (PC Make/Reflux Ratio), which are important in the models.

6.2. Effect of the softplus transformation

The softplus transformation guarantees positive predictions for the impurity while avoiding unintended nonlinear effect since it is asymptotically linear for large values. The overall performances of the softplus transform for the two models are compared in Table 3. It is observed that the use of softplus transformation not only guar-

Table 3

R^2 and Q^2 of the LARS and PLS models with or without softplus transformation.

	LARS	LARS	PLS	PLS
	w/o softplus	w/ softplus	w/o softplus	w/ softplus
R^2	0.813	0.825	0.858	0.864
Q^2	0.622	0.682	0.492	0.499

Table 4

R^2 and Q^2 of the LARS and PLS models with real data only or with interpolated data for training.

	LARS with interpolated data	LARS with real data	PLS with interpolated data	PLS with real data
R^2	0.816	0.813	0.860	0.858
Q^2	0.590	0.622	0.494	0.492

antees positive impurity prediction, it also improves the model accuracy marginally.

6.3. Effect of interpolated impurity data

The Dow datasets have hourly impurity data that were linearly interpolated to match the sampling rate of the process data. To compare the effect of interpolated data on modeling, the R^2 and Q^2 of the LARS and PLS models trained with or without using the interpolated data are shown in Table 4. It is observed from the results that the use of interpolated data does not improve the model accuracy for either the training dataset or the validation dataset. Therefore, the interpolated data do not provide more information to obtain a better model. However, the models built in this work can be used to predict the inter-sample impurity based on process measurements.

7. Conclusions

An integrated procedure of statistical machine learning and process knowledge is proposed for the Dow data challenge problem, which successfully addressed several challenges in the problem. The solution procedure is able to select important variables that are confirmed with process knowledge. The softplus

transform applied to the impurity successfully avoided negative predictions from the model and unintended nonlinear distortions. Process knowledge plays an important role from exploratory data analysis to final model interpretation. Finally, the online bias learning based on available data is effective in handling the process drift and improves the prediction accuracy significantly.

In this paper, partial least squares and least angle regression are demonstrated to have complementary strengths due to their different modeling philosophies. PLS uses all predictor variables and deals with collinearity effectively, while LARS performs a greedy selection of relevant variables but tends to ignore collinear variables. The prediction errors of both models are comparable, especially with the online bias learning implemented. However, if the models are used for control purposes, the PLS model that includes physically collinear variables could be preferred.

We finally note that variable selection can be performed for PLS models (Hoskuldsson, 2001; Anzanello and Fogliatto, 2014; Wang et al., 2015; Chun and Keleş, 2010), which is a departure from early views in the PLS community (Baroni et al., 1992; Kresta et al., 1994; Wold et al., 1996, 1998). The issue of variable selection for PLS models deserves further studies, since it is beneficial to include all relevant and collinear variables in the model. Selecting only one out of a collinear group of variables like what LARS does in this work could fail to capture inherent process collinearity.

Author Contribution Statement

All authors have read this revised manuscript and approved its resubmission.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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